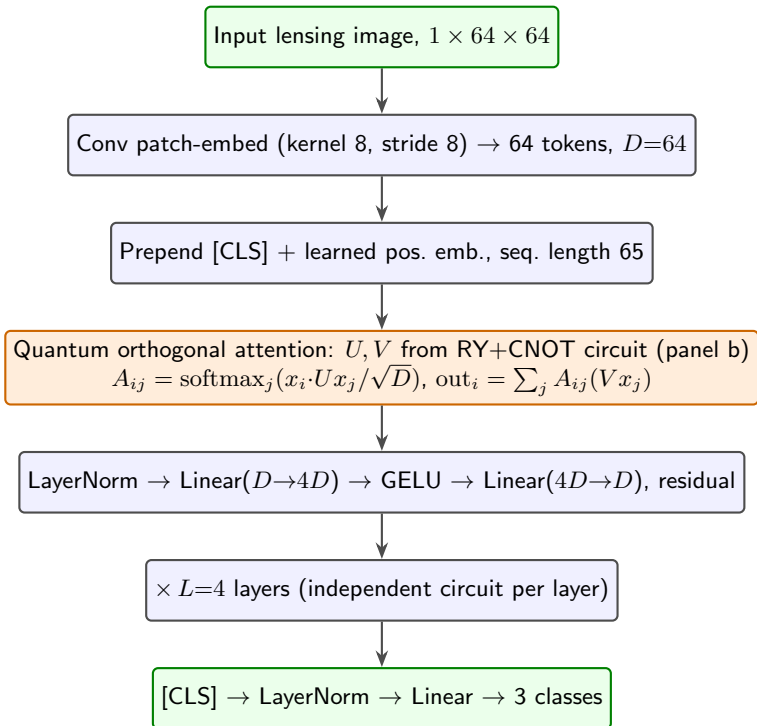
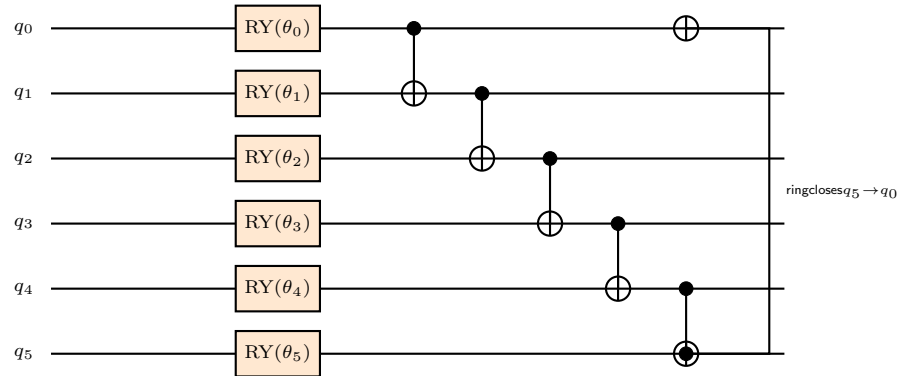


(a) QOVT pipeline (quantum arm)



(b) RY+CNOT ring circuit (one of $L_{\text{circ}}=8$ layers)



$$T = L_{\text{circ}} \times N_Q \times 2 \text{ matrices} \times L = 8 \times 6 \times 2 \times 4 = 384 \text{ total angles}$$

repeat this layer $8 \times$ (independent θ each time) $\rightarrow$ one 64×64 orthogonal matrix U